

Adversarial Co-Evolution of RL and LLM Agents in Gin Rummy: A Phase-1/2 System Report

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Abstract

We build a hybrid system in which a lightweight action-masked PPO agent and a strong-but-slow LLM opponent train against each other in Gin Rummy, without paying full LLM inference latency on every reinforcement-learning step. This report covers two phases. In **Phase 1** an action-masked PPO agent saturates against a random opponent: an eight-configuration hyperparameter sweep lands every run in a 98.3%–99.6% win-rate band, statistically indistinguishable at 1000-episode evaluation. Because further tuning against weak opponents is unproductive, **Phase 2** builds the infrastructure for LLM-in-the-loop training: a master/worker/cache inference stack with shared-filesystem service discovery and a suit-symmetry prompt cache, plus self-play and AlphaZero-style pool curricula for an in-distribution opponent. We report three operational findings: (i) OLMoE-1B is too weak to teach (it plays at chance), whereas Qwen2.5-7B-Instruct beats a random hero 5/5 and is a genuinely useful teacher; (ii) loading a 7B worker from the home NFS is $62\times$ slower than from scratch/BeeGFS and must be avoided; and (iii) pool self-play diverged after $\sim 10\text{M}$ steps, dropping its win rate vs random from 98.5% to 85.8%. Finally we run a full **1.5M-step RL-vs-LLM fine-tune** (PPO warm-started from the self-play champion, up to 32 Qwen2.5-7B workers): episode reward stays flat at the knock value and the resulting agent matches—but does not beat—its progenitor. The honest conclusion is that a 7B opponent is *weaker* than our self-play champion, so it cannot lift a near-saturated policy; making the LLM a stronger or co-evolving teacher is the Phase-3 problem.

1 Goal and roadmap

Gin Rummy is an imperfect-information card game requiring both short-horizon arithmetic (deadwood counting) and long-horizon planning (meld formation). Our roadmap has three phases: (1) train a strong RL backbone against weak opponents; (2) train that backbone against an LLM opponent that is strategically richer but operationally far slower; (3) let the two co-evolve. We use action-masked PPO [4, 1] on the PettingZoo [5] `gin_rummy_v4` environment (wrapping RLCard [6]) with Stable-Baselines3 [3]; masking sets illegal-action logits to $-\infty$ inside a custom actor-critic policy. We first fixed an upstream bug in which PettingZoo silently reverts the configured `knock/gin` rewards on every seeded reset (PettingZoo PR #1312); all results below use the corrected rewards (`knock` = 0.5, `gin` = 1.5).

2 Phase 1: the RL backbone saturates against random

We sweep eight configurations: a 3×2 grid over learning rate $\{3\cdot 10^{-4}, 10^{-4}, 5\cdot 10^{-5}\}$ and entropy coefficient $\{0.01, 0.03\}$, plus two ablations (rollout length 1024; 10 epochs). Each trains for 2×10^6 steps with 96 parallel envs on a 64-core node and is evaluated over 1000 deterministic games vs **RandomAgent**. Every configuration converges to the 98.3%–99.6% band (Fig. 1, Table 1); the spread is about one binomial 95% CI (± 0.6 pp at $p=0.99$), so the configurations cannot be separated. The shared failure mode is informative: mean episode reward clusters at the knock value (0.5) rather than the gin value (1.5) (Fig. 1b), i.e. the agent takes the certain knock instead of holding for a gin—a risk/reward decision that only a *thinking* opponent can teach. This is what motivates Phase 2.

3 Phase 2: infrastructure for LLM-in-the-loop training

A single PPO rollout fires up to $\sim 50\text{k}$ opponent queries; at 0.5–3 s per 7B call, a naïve loop takes hours. We decouple inference from RL with a three-tier stack. **Workers** (one GPU each, HF backend) load the model and register themselves in a shared-filesystem registry. The **master** (CPU) discovers live workers, load-balances, and owns a prompt cache. The RL **client** (96 env subprocesses) reaches the master through the existing Ollama-compatible API, so no RL code changes. We chose the HuggingFace backend because the installed vLLM [2] 0.8.5 is incompatible with transformers 5.7. The cache uses a *suit-symmetry canonical key*: Gin Rummy is invariant under suit relabeling, so suit-equivalent states share one entry and the chosen action is de-canonicalized on return.

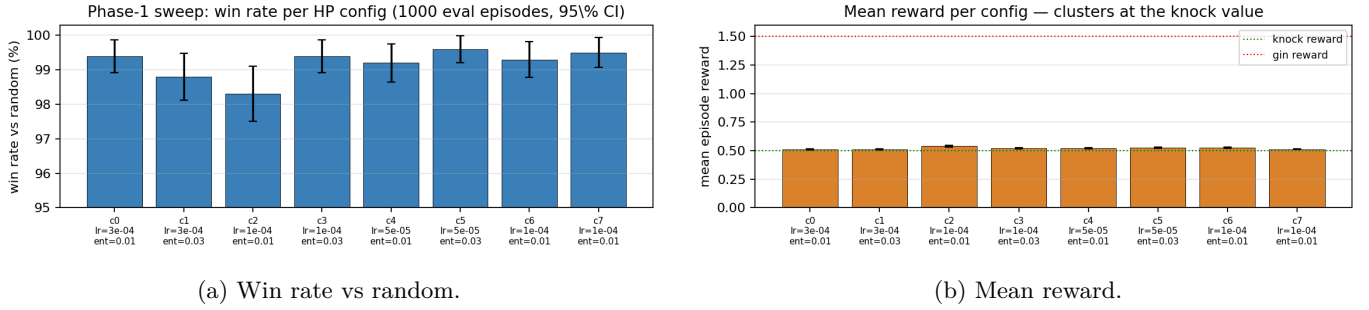


Figure 1: Phase-1 sweep. All eight configs saturate vs random (a); reward sits at the knock value, far below gin (b), exposing the improvement headroom.

Table 1: Phase-1 leaderboard (top rows) and Phase-2 opponent panel (bottom). Win/loss over 1000 deterministic eval games unless noted.

Agent / config	opponent	win%	loss%	note
PPO cfg5 (lr 5e-5, ent .03)	random	99.6	0.4	best of sweep
PPO cfg7 (lr 1e-4, 10 epochs)	random	99.5	0.5	ablation
PPO cfg0 (lr 3e-4, ent .01)	random	99.4	0.6	baseline
PPO cfg3 (lr 1e-4, ent .03)	random	99.4	0.6	
PPO cfg6 (lr 1e-4, $n_s=1024$)	random	99.3	0.7	ablation
PPO cfg4 (lr 5e-5, ent .01)	random	99.2	0.8	
PPO cfg1 (lr 3e-4, ent .03)	random	98.8	1.2	
PPO cfg2 (lr 1e-4, ent .01)	random	98.3	1.7	worst of sweep
Self-play (3M, vs frozen run_5)	run_5	61.1	38.9	beats progenitor
Self-play (3M)	random	98.7	1.3	stays dominant
Pool champion (12M)	random	85.8	14.2	<i>diverged</i>
Pool champion (12M)	run_5	62.0	38.0	
OLMoE-1B opponent	random	≈ 50	—	plays at chance
Qwen2.5-7B opponent	random	100	0	5/5 (small N)
LLM-finetuned PPO (1.5M)	random	99.0	1.0	retains competence
LLM-finetuned PPO (1.5M)	self-play champ	43.3	56.7	<i>no net gain</i>
LLM-finetuned PPO (1.5M)	winrate / reward	55.3 / 61.3	—	still beats weak agents

Two infrastructure findings. First, loading a 7B worker from the home NFS runs at ~ 11 MB/s and takes ~ 28 min—long enough to blow the health-check timeout. Staging weights on scratch/BeeGFS cuts this to 27 s, a $62\times$ speedup (Fig. 3c); this is mandatory when launching dozens of workers. Second, the in-game opponent model is selected purely by the worker’s loaded weights—the master ignores the client’s model string—so swapping opponents is a one-line preset change.

4 Opponents: self-play, pool, and the LLM

Self-play. Fine-tuning the Phase-1 champion against a *frozen* copy of itself for 3M steps yields 58–61% vs that progenitor while holding $\sim 98.7\%$ vs random (Fig. 3a)—one clean generation of improvement.

Pool (AlphaZero-style). Training against a growing pool of past checkpoints ran to 12M steps but *diverged* after ~ 10 M: vs-random win rate fell from $\sim 98.5\%$ to 85.8% (Fig. 3b). We keep the pre-divergence checkpoints; the production opponent reverts to self-play.

LLM opponent. OLMoE-1B cannot follow the action-format prompt—most responses fail to parse and fall back to a random move, so it plays at chance and cannot teach. Qwen2.5-7B-Instruct, by contrast, beats a random hero 5/5 (Fig. 3d): a genuinely competent teacher. Its chain-of-thought prompt costs ~ 16 s/call (1024 output tokens); for training we switch to a terse prompt that emits only the action and cap output tokens, cutting latency by an order of magnitude.

Human play. A zero-dependency web client (HTML/JS, Fig. 2) lets a human play any of the four trained opponents (Phase-1, self-play, pool, reward-shaped) with 3-D card animations; it reads opponent state from RLCard and was validated in headless Chromium.



Figure 2: The human-vs-RL web client (debug view, opponent hand revealed). The player drafts and discards against any trained agent; the board animates draws, reveals, and discards. The same **LLMagent** path that the RL loop uses also drives an optional LLM opponent here.

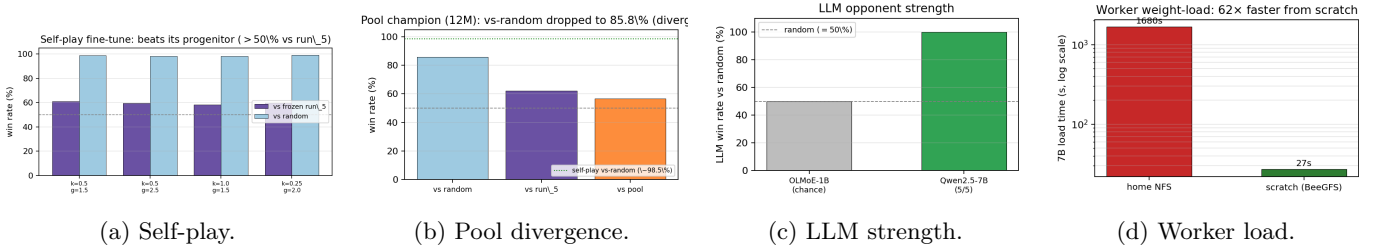


Figure 3: Phase-2 results. (a) self-play beats its progenitor; (b) the pool champion regressed vs random after divergence; (c) Qwen2.5-7B is a competent opponent where OLMoE-1B is not; (d) scratch staging is 62 $\times$ faster.

5 Full RL-vs-LLM run: results, limitations, next steps

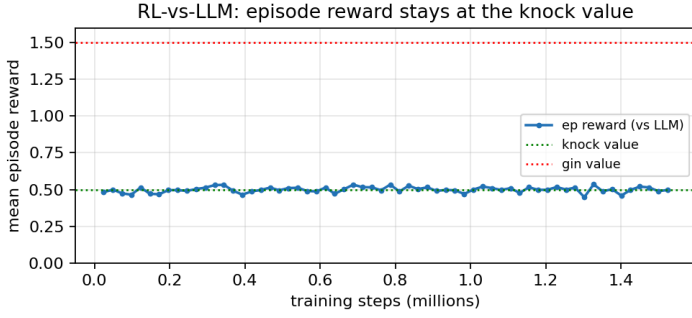
We then ran the real experiment: PPO warm-started from the self-play champion, 96 env subprocesses, fine-tuned for **1.5M steps** against the master + a pool of up to 32 Qwen2.5-7B workers (the scheduler granted ~ 14 under the shared GPU quota; the pool is elastic). The serving stack is the headline infrastructure win: the terse prompt + a 64-token cap cut effective per-call latency from ~ 16 s to ~ 0.4 s, the master sustained ~ 32 load-balanced queries/s ($\sim 500\times$ a single-stream chain-of-thought loop), and PPO held ~ 39 env-steps/s—so a rollout completes in minutes and the whole run finished in a few hours.

The learning result is negative and clean. Across all 1.5M steps the mean episode reward stays pinned at the knock value ≈ 0.49 and never trends toward gin (1.5) (Fig. 4a). The fine-tuned agent *retains* 99.0% vs random and still beats the weaker reward-shaped agents, but sits at 43.3% vs the self-play champion it started from (Fig. 4b, Table 1) —no net gain. The cause is structural: Qwen2.5-7B with a terse prompt is a *weaker* player than our self-play champion, so it provides no gradient toward stronger play; training a near-saturated policy against a weaker teacher holds it in place (and slightly perturbs it). The suit-symmetry cache also stayed near 0% hit-rate—opening states are too diverse to recur at this scale—so throughput is purely worker-bound.

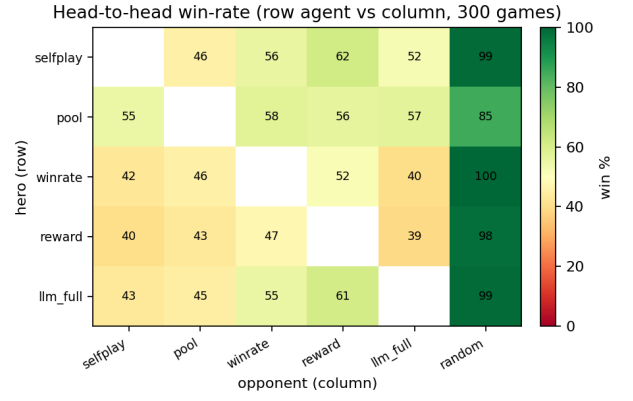
Limitations. (i) The teacher is weaker than the trainee—the central reason no improvement appears; (ii) the terse prompt trades opponent quality for throughput, and we have not measured a strong-prompt / larger-model teacher; (iii) single seed per run and a small ($N=5$) LLM-strength sample; (iv) the prompt cache yields no speedup at current volume; (v) pool self-play diverged past 10M; (vi) the midterm’s negative Dagger and dense-reward-shaping results remain code-unaudited.

Next steps. (1) *Co-evolution* (Phase 3): make the LLM a stronger, *evolving* teacher—seed its few-shot prompt with the RL agent’s strongest games and/or use a larger model (e.g. Qwen2.5-32B), then alternate generations so both sides improve. (2) Reward-shape directly toward the headroom (penalize early knocks / bonus held gins), since the gin decision is exactly what Phase 1 left on the table. (3) Offline distillation of bulk LLM-vs-LLM games with CQL. (4) Three-seed confirmation and a strong- vs terse-prompt teacher ablation.

Reproducibility. Sweeps run as SLURM arrays under `sweep/`; the LLM stack is three `slurm/` jobs — master, a worker array (scratch-staged weights), and the trainer; every figure here is regenerated by `make_figures.py` from the JSON result files.



(a) Reward flat over 1.5M steps.



(b) Head-to-head win-rate.

Figure 4: Full RL-vs-LLM run. (a) episode reward never leaves the knock value, so the LLM teacher did not raise the gin rate; (b) the fine-tuned agent (`llm_full`) lands at champion level—beating the weak agents and random, roughly even vs the champion it came from.

References

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